RAVM-DT Supplementary Protocol and Results

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This supplement accompanies *Joint Evidence Gating for Continual Maintenance of Predictive Digital Twin Models Under Unreliable Monitoring*. It documents the executed protocol and existing results, including adverse outcomes. Tables use the frozen source-domain risk thresholds unless explicitly stated. Source identities and scripts are supplied as separate machine-readable attachments. Table reconstruction is not an independent repetition of training.

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S1 Synthetic Forecasting Task and Generator

The target $y_j \in [0, 1]$ is a latent workload. Each input has twelve monitored metrics followed by twelve masks. At base resolution the logit is a sampled level plus seasonal components with periods 24, 96, and 672, respective amplitude multipliers $0.35, 0.75, 0.25$, and independent uniform phases; an AR(1) process; and positive Gaussian-shaped bursts. AR innovations have the type-specific Gaussian standard deviation. The number of bursts is Poisson with mean (sampled rate)$\times T_{base}/1000$; centers are uniform in the base record and widths uniform on $[2, 14]$. The sigmoid clips logits to $[-12, 12]$ before transforming. Base random drift probabilities are set to zero for the registered four-regime task, so they do not add an uncontrolled second drift mechanism.

Centers use a prefix-balanced, randomly shuffled A:B:C schedule of 3:4:3 in each complete ten-center block. Main and new supplemental pools use generator seed 7311 and offset $17c_0$, where c_0 is the first center identifier. A center-specific random-state seed comes from this pool state. The simulation returns 17,000 observations per

center, sampled at its own integer spacing Δ from the base-resolution signal; indices are $8 + j\Delta$. Thus sampling heterogeneity is real subsampling, not a feature tag. Complete numeric type ranges are in Table S1.

Table S1: Executed synthetic type ranges; continuous parameters are uniform within their ranges. Integer counts are discrete, with endpoints included where configured.

Parameter	Type A	Type B	Type C
Sampling spacing	1	1, 2	2, 3, 4
Point missing probability	.02–.08	.10–.25	.25–.45
Nominal missing block length	5–15	20–60	60–120
Observation noise SD	.01–.03	.03–.06	.06–.10
Dropped schema channels	0–2	2–4	4–6
Base logit	-.70–.10	-.45–.35	-.20–.55
Seasonal amplitude	.20–.45	.30–.60	.40–.75
AR coefficient	.55–.80	.60–.88	.65–.92
Process SD	.04–.08	.06–.11	.08–.14
Burst rate per 1,000	1–2.5	1.5–3.5	2–4.5
Burst amplitude	.25–.55	.35–.75	.45–.95

Metric k transforms the delayed target logit using gain $g_k \in [.65, 1.45]$, bias in $[-.35, .35]$, a quadratic term $q_k(y-.5)^2$ with $q_k \in [-.20, .35]$, an amplitude-.08 daily sinusoid, and Gaussian local noise with SD in $[.01, .04]$, followed by a sigmoid. Metric delay is a uniform integer 0–8 base steps. Type-specific Gaussian observation noise is then added, values are clipped to $[0, 1]$, and missing entries are zeroed with explicit masks. Point missingness affects only available schema channels. Each center adds 2–6 single-metric outages with length drawn from $[\max(1, \lfloor b/3 \rfloor), b]$, 1–3 grouped outages affecting 2–4 available metrics with length $[\max(2, \lfloor b/4 \rfloor), \lfloor b/2 \rfloor]$, and, with probability .20, 0–2 full outages of length 2–8. Here b is the sampled nominal missing-block length. Permanent schema drops are reapplied last. Latent targets are never generated from masked observations.

Let j index the returned observation, starting at zero. The additional regime shift is $q_j = 0$ (stationary), $.8\mathbf{1}[j \geq 6520]$ (abrupt), $.8\,\mathrm{clip}((j - 6520)/5120, 0, 1)$ (gradual), or $.8\mathbf{1}[j \in [5240, 9080) \cup [12920, 17000)]$ (recurring). It is added in target-logit space and, with the metric's sampled gain and integer delay divided by Δ, in metric-logit space. Types and regimes are distinct: regimes cycle by center order. The attached `d3_engine.py`, `datasets.py`, `vendor/main_cfg.py`, and `vendor/sim.py` give the exact rounding and random-number ordering.

S2 Temporal Roles, Models, Development, and Fallback

For workload data the common warm-up is $[0, 800)$ and initial historical Verification is $[800, 1000)$. At opportunity $t \in \{1, \ldots, 24\}$, $v_t = 1400 + (t-1)640$: Update is $[v_t - 400, v_t)$, Verification $[v_t, v_t + 240)$, Service $[v_t + 240, v_t + 640)$. Lookback $L = 96$, horizon $H \in \{1, 4\}$, and every target horizon stays inside its designated role. A role supplies input windows and subsequent H workload targets; purges between blocks follow the supplied `windows` and `block_indices` functions. Recent Verification has six chronological blocks; the rank routine pools their valid window losses. Historical controls retain at most twelve blocks. Native legacy gates use their own frozen five-block robust implementation instead of being substituted with the new score.

Workload sources were pretrained before target development on centers 10000–10019, raw source segments $[0, 1000)$, separately by architecture/horizon/seed. The supplied original pretraining script uses MSE, Adam with learning rate .001, 50 epochs, batch size 64, per-epoch seeded permutations, and model seeds 7311–7313. Historical sidecars record epochs, source centers, source-state hashes, and segment identities. The corresponding source checkpoint bytes are indexed rather than replaced with a newly fitted model. A target center/seed initializes all methods from the same 50-step warm-up state. Each subsequent method maintains its own causal deployed chain. Source pretraining, shared target warm-up, and candidate adaptation are different operations.

GRU uses one recurrent layer with 32 hidden units and a linear $32 \to H$ head. MLP flattens $96F$ features into a 64-unit ReLU layer and an H-output linear head. TCN uses two causal convolution layers with 32 channels, kernel 3, dilations 1 and 2, ReLU activation, right-padding truncation 2 and 4, and a $32 \to H$ last-state head. Workload $F = 24$; UCI $F = 40$; ThermBuild $F = 26$. Candidate adaptation uses 50 SGD steps at learning rate .01, replacement batches of size $\min(64, n)$ and norm clipping 1. Training and evaluation switch model mode explicitly.

Native ER/EWC/Huber policies retain their implemented generators. ER capacity is 24; EWC coefficient is 10, Fisher decay .9 and at most 32 Fisher examples; Huber delta is .1. The original robust recent gate has $B = 5$, $\beta = .1$, recent dispersion coefficient .5, historical coefficient 0, improvement margin $2.8018237766243602 \times 10^{-5}$ and retention margin 0.

Joint and neighboring gate development. Write $R = 0.00560364755324872$ for the development loss scale. Joint weights are $\{0, .05, .15, .3\}$ and margin fractions of R are $\{.01, .05, .1, .2\}$, equivalent to $\{1, 5, 10, 20\}\eta$. Each RecentMean, MWU-Mean, and DualMean control searches fractions $\{0, .001, .002, .003, .005, .0075, .01, .015, .02, .03, .04, .05, .075, .1, .15, .2\}R$. Two fixed historical controls complete 66 arms. DevCal has eight new centers, one seed (7311), three streams and 24 stages per arm. The gate-selection rule first requires per-stream harmful frequency $\leq .05$ and downside $\leq \eta$, with coherent-sticky frequency additionally $\leq .045$. Among active positive-weight eligible Joint settings, it minimizes clean/mixed balanced MSE, with exact ties broken by lower measured maintenance time, smaller weight, larger margin, then identifier. Eleven of sixteen Joint-grid cells pass risk eligibility; nine are active and have positive weight. This is distinct from Pareto membership. The selected rule is $d_t < 0$ and $d_t + .3 \max(\bar{\ell}_t^P, 10^{-12})(p_t - .2) \leq -5.6036475532487204 \times 10^{-5}$; p_t is the frozen one-sided asymptotic MWU output, not a calibrated future-risk probability.

Neighboring controls first choose eligible active arms, then qualified inactive arms; if none is qualified, the minimum-MSE arm is only descriptive and cannot serve as a qualified active comparator. Final margins are $4.20273566493654 \times 10^{-4}$ for RecentMean and DualMean, and $2.8018237766243602 \times 10^{-5}$ for MWU-Mean. Development and correction locks precede the corrected Main Final. Main's primary comparator is subsequently the lowest observed Final MSE among all eleven qualified active alternatives; its selected-comparator uncertainty is not accounted for by the pairwise intervals. This choice is different from the prelocked comparator of the earlier stopped Confirm.

Table S2: Original ValidationCommit calibration, not the new Main evaluation: 1,440 opportunities per threshold/stream. None qualifies in both streams. Thresholds are multiples of the original R scale.

Threshold / R	Stream	Commits	Harmful	Harm (%)	Qualified stream
0	clean	869	142	9.861	False
0	mixed	1054	363	25.208	False
0.001	clean	804	132	9.167	False
0.001	mixed	1038	363	25.208	False
0.005	clean	637	89	6.181	False
0.005	mixed	962	337	23.403	False
0.01	clean	517	69	4.792	True
0.01	mixed	845	284	19.722	False
0.05	clean	227	14	0.972	True
0.05	mixed	419	123	8.542	False

The historical calibration reconstructed 600 trajectories and 14,400 threshold/stream/stage records. At threshold zero, commits are 60.347%/73.194% and harmful frequencies 9.861%/25.208% in clean/mixed. At the largest tested margin, mixed harmful frequency remains 8.542%. The native risk-constrained fallback therefore retains all states; it is recorded as `validation_retain_all=true`, threshold 0 in the new Main configuration. This is neither proof that validation can never adapt nor a matched search against Joint's later development. Active neighboring controls address that weakness without retrospectively loosening this fallback.

S3 Monitoring Interventions

Faults change maintenance-role copies of input metrics and, for specified mechanisms, observed training/Verification targets. They do not modify the held-out Service reference. Main clean has zero intervention; Mixed uses nominal fraction .10. In each four-stage cycle a seeded permutation assigns random, persistent-bias, burst, and replay mechanisms, common to all model methods. In each of ten chronological raw-observation strata b, at most $\lfloor \alpha n_b \rfloor$ entries are selected without replacement, except burst chooses one contiguous run of that length per stratum. The masks depend on seed 9417, center, stage and role, independently of model seed or method.

Random changes affect all available metric channels and observed targets with Gaussian SD .15 clipped to $\pm .2$. Bias and burst use a common stage-specific random sign times .2 on a randomly chosen half of metric channels (at

least one) and on the selected targets. Metric changes respect observation masks and clip normalized values to $[0, 1]$; target values are clipped likewise. Replay copies selected full feature/mask rows and targets from the corresponding positions of the final n observations of the old warm-up segment ending at index 800. Source timestamps must precede the attacked segment. No future interpolation supplies replay values.

Coherent-sticky instead fixes a channel and sign by center: compute SHA-256 of the UTF-8 string `B1-STICKY:9417:`*center*; the first four digest bytes interpreted little-endian modulo 12 give the channel, and the fifth byte's parity gives the sign (even positive). The same signed .2 shift affects this channel, conditional on its presence mask, and the observed target at all stages 1–24. Each role samples $\lfloor .10n \rfloor$ positions uniformly without replacement using SeedSequence $[9417, c, t, r, 51]$, where $r = 0$ for Update, 1 for Verification. It is a globally fixed-fault direction, not the ten-stratum Mixed mask. Service and all observation masks remain unchanged.

E1 uses $\alpha = .05, .10, .20$: Mixed acts on both Update and Verification, whereas persistent-bias and replay stress streams act on Verification only. Their curves do not isolate fault type independently of fault location. For UCI, seed 9487 and room/stage/role choose 10% within each of ten strata; bias adds 2°C to the selected room metric and target; dropout zeros its input and observation mask without changing the target; stale copies the room input and observed target from six earlier positions within the segment (clipped at position zero). ThermBuild uses one contiguous four-row (one-hour) episode per 192-row Update and Verification segment, a 2°C target-channel/target offset or target-channel/mask dropout, seeded by 27013, building, pool origin, raw stage, and role. These numerical settings are controlled stresses, not estimated field rates. The sensor-fault and networked-observation literature supports testing these mechanism classes, not the chosen numerical ranges [26, 27].

S4 Complete Supplemental Outcome Tables

All registered methods are retained below, including ineligible, inactive, and unfavorable comparators. MSE is in each task's normalized squared units; harm and commitment use all opportunities. Downside is $N^{-1} \sum a_t \max(z_t, 0)$; harmful frequency is $h = N^{-1} \sum a_t \mathbf{1}[z_t > 0]$. This h is distinct from forecast horizon H. The qualification column tests both $h \leq .05$ and downside $\leq \eta$ separately per stream. Beneficial recall and all comparator intervals are additionally supplied in the CSVs; seed/center macro recall and count-based micro recall have different aggregation weights and must not be interchanged.

Table S3: E1: all eight methods in nine stress streams; 24 centers, three seeds, 24 stages. Labels a005/a010/a020 denote 5/10/20 percent corruption.

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
Joint Gate	mixed_a005	3.29615	35/1728	2.025	2.12185	19.850	Yes
Joint Gate	mixed_a010	3.34113	78/1728	4.514	8.96686	21.354	Yes
Joint Gate	mixed_a020	3.42120	147/1728	8.507	50.40665	25.868	No
Joint Gate	persistent_bias_a005	3.29533	19/1728	1.100	1.20093	16.725	Yes
Joint Gate	persistent_bias_a010	3.33940	14/1728	0.810	0.96221	14.931	Yes
Joint Gate	persistent_bias_a020	3.39663	9/1728	0.521	0.76976	12.326	Yes
Joint Gate	replay_a005	3.30355	15/1728	0.868	0.88882	15.625	Yes
Joint Gate	replay_a010	3.32386	8/1728	0.463	0.78889	14.005	Yes
Joint Gate	replay_a020	3.39625	5/1728	0.289	0.57722	11.400	Yes
RecentMean	mixed_a005	3.31266	16/1728	0.926	1.74183	15.394	Yes
RecentMean	mixed_a010	3.29906	91/1728	5.266	12.99364	22.454	No
RecentMean	mixed_a020	3.43208	356/1728	20.602	116.56318	46.586	No
RecentMean	persistent_bias_a005	3.30177	10/1728	0.579	0.81485	13.773	Yes
RecentMean	persistent_bias_a010	3.28442	14/1728	0.810	0.98210	15.336	Yes
RecentMean	persistent_bias_a020	3.28941	33/1728	1.910	2.10323	19.271	Yes
RecentMean	replay_a005	3.34882	12/1728	0.694	1.08951	11.921	Yes
RecentMean	replay_a010	3.36432	9/1728	0.521	0.79738	11.400	Yes
RecentMean	replay_a020	3.42655	7/1728	0.405	0.75160	10.417	Yes
MWU-Mean	mixed_a005	3.51525	6/1728	0.347	0.45493	10.475	Yes
MWU-Mean	mixed_a010	3.57269	22/1728	1.273	1.87607	11.053	Yes
MWU-Mean	mixed_a020	3.59864	23/1728	1.331	6.77408	10.648	Yes
MWU-Mean	persistent_bias_a005	3.49601	3/1728	0.174	0.17869	9.606	Yes
MWU-Mean	persistent_bias_a010	3.54912	2/1728	0.116	0.13238	8.333	Yes
MWU-Mean	persistent_bias_a020	3.60976	0/1728	0.000	0.00000	6.829	Yes
MWU-Mean	replay_a005	3.48346	4/1728	0.231	0.10571	9.896	Yes
MWU-Mean	replay_a010	3.51880	4/1728	0.231	0.30852	8.970	Yes
MWU-Mean	replay_a020	3.57332	0/1728	0.000	0.00000	7.581	Yes
DualMean	mixed_a005	3.40013	9/1728	0.521	0.61101	13.137	Yes
DualMean	mixed_a010	3.41649	37/1728	2.141	3.69002	15.336	Yes
DualMean	mixed_a020	3.55862	74/1728	4.282	16.76936	17.072	Yes
DualMean	persistent_bias_a005	3.37699	7/1728	0.405	0.60604	11.979	Yes
DualMean	persistent_bias_a010	3.38104	13/1728	0.752	0.89698	13.426	Yes
DualMean	persistent_bias_a020	3.45818	27/1728	1.562	1.91660	15.046	Yes

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
DualMean	replay_a005	3.42987	6/1728	0.347	0.53632	10.822	Yes
DualMean	replay_a010	3.46195	5/1728	0.289	0.47571	9.954	Yes
DualMean	replay_a020	3.52679	6/1728	0.347	0.45095	9.722	Yes
Original RAVM	mixed_a005	3.39930	63/1728	3.646	2.05479	26.100	Yes
Original RAVM	mixed_a010	3.46725	56/1728	3.241	3.20589	23.843	Yes
Original RAVM	mixed_a020	3.62446	52/1728	3.009	6.55758	20.660	Yes
Original RAVM	persistent_bias_a005	3.36858	33/1728	1.910	1.54149	26.157	Yes
Original RAVM	persistent_bias_a010	3.38032	44/1728	2.546	2.22263	26.273	Yes
Original RAVM	persistent_bias_a020	3.48861	39/1728	2.257	2.73802	24.653	Yes
Original RAVM	replay_a005	3.37472	34/1728	1.968	1.23131	25.637	Yes
Original RAVM	replay_a010	3.40483	29/1728	1.678	1.25375	23.148	Yes
Original RAVM	replay_a020	3.48250	24/1728	1.389	2.04637	21.412	Yes
Always-FT	mixed_a005	3.05698	578/1728	33.449	74.51068	100.000	No
Always-FT	mixed_a010	3.14763	710/1728	41.088	132.42845	100.000	No
Always-FT	mixed_a020	3.48070	813/1728	47.049	341.02852	100.000	No
Always-FT	persistent_bias_a005	2.99452	427/1728	24.711	42.64905	100.000	No
Always-FT	persistent_bias_a010	2.99452	427/1728	24.711	42.64905	100.000	No
Always-FT	persistent_bias_a020	2.99452	427/1728	24.711	42.64905	100.000	No
Always-FT	replay_a005	2.99452	427/1728	24.711	42.64905	100.000	No
Always-FT	replay_a010	2.99452	427/1728	24.711	42.64905	100.000	No
Always-FT	replay_a020	2.99452	427/1728	24.711	42.64905	100.000	No
Huber-FT	mixed_a005	3.26089	463/1728	26.794	41.39826	100.000	No
Huber-FT	mixed_a010	3.30690	604/1728	34.954	64.26036	100.000	No
Huber-FT	mixed_a020	3.47049	716/1728	41.435	166.25342	100.000	No
Huber-FT	persistent_bias_a005	3.22876	415/1728	24.016	32.43374	100.000	No
Huber-FT	persistent_bias_a010	3.22876	415/1728	24.016	32.43374	100.000	No
Huber-FT	persistent_bias_a020	3.22876	415/1728	24.016	32.43374	100.000	No
Huber-FT	replay_a005	3.22876	415/1728	24.016	32.43374	100.000	No
Huber-FT	replay_a010	3.22876	415/1728	24.016	32.43374	100.000	No
Huber-FT	replay_a020	3.22876	415/1728	24.016	32.43374	100.000	No
Frozen	mixed_a005	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	mixed_a010	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	mixed_a020	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	persistent_bias_a005	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	persistent_bias_a010	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	persistent_bias_a020	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	replay_a005	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	replay_a010	5.52676	0/1728	0.000	0.00000	0.000	Yes
Frozen	replay_a020	5.52676	0/1728	0.000	0.00000	0.000	Yes

Table S4: E2: all six methods in every architecture/horizon and stream; 24 centers, three seeds, 24 stages per group.

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
Joint Gate	GRU_H1/clean	3.14048	15/1728	0.868	0.97612	17.419	Yes
Joint Gate	GRU_H1/mixed	3.20394	75/1728	4.340	9.39979	21.875	Yes
RecentMean	GRU_H1/clean	3.18799	3/1728	0.174	0.06752	12.616	Yes
RecentMean	GRU_H1/mixed	3.15874	81/1728	4.688	11.54566	21.412	Yes
MWU-Mean	GRU_H1/clean	3.34133	1/1728	0.058	0.02338	10.475	Yes
MWU-Mean	GRU_H1/mixed	3.40038	12/1728	0.694	0.81665	11.053	Yes
DualMean	GRU_H1/clean	3.27621	1/1728	0.058	0.01629	10.995	Yes
DualMean	GRU_H1/mixed	3.30489	32/1728	1.852	3.94075	14.583	Yes
Original RAVM	GRU_H1/clean	3.26014	27/1728	1.562	1.02449	24.595	Yes
Original RAVM	GRU_H1/mixed	3.34544	59/1728	3.414	3.35859	23.264	Yes
Always-FT	GRU_H1/clean	2.83827	423/1728	24.479	33.90472	100.000	No
Always-FT	GRU_H1/mixed	2.96865	691/1728	39.988	114.98148	100.000	No
Joint Gate	GRU_H4/clean	3.74585	45/1728	2.604	3.57148	24.653	Yes
Joint Gate	GRU_H4/mixed	3.84080	94/1728	5.440	12.07288	26.389	No
RecentMean	GRU_H4/clean	3.81552	17/1728	0.984	1.85203	17.535	Yes
RecentMean	GRU_H4/mixed	3.81460	113/1728	6.539	16.38288	27.488	No
MWU-Mean	GRU_H4/clean	4.01672	22/1728	1.273	1.65036	16.667	Yes
MWU-Mean	GRU_H4/mixed	4.10124	37/1728	2.141	4.84796	16.030	Yes
DualMean	GRU_H4/clean	3.95818	7/1728	0.405	0.84345	14.641	Yes
DualMean	GRU_H4/mixed	4.01825	45/1728	2.604	6.07429	18.287	Yes
Original RAVM	GRU_H4/clean	4.12968	27/1728	1.562	1.98785	22.049	Yes
Original RAVM	GRU_H4/mixed	4.23825	52/1728	3.009	5.61306	21.296	Yes
Always-FT	GRU_H4/clean	3.46800	443/1728	25.637	49.05201	100.000	No
Always-FT	GRU_H4/mixed	3.59690	655/1728	37.905	124.78377	100.000	No
Joint Gate	MLP_H1/clean	5.04326	42/1728	2.431	6.24662	21.065	Yes
Joint Gate	MLP_H1/mixed	5.09264	64/1728	3.704	13.71736	20.775	Yes
RecentMean	MLP_H1/clean	5.07287	19/1728	1.100	4.59097	17.419	Yes
RecentMean	MLP_H1/mixed	5.05653	83/1728	4.803	19.85343	24.016	Yes
MWU-Mean	MLP_H1/clean	5.16642	10/1728	0.579	2.70019	13.773	Yes
MWU-Mean	MLP_H1/mixed	5.20126	25/1728	1.447	5.86388	14.815	Yes
DualMean	MLP_H1/clean	5.22686	14/1728	0.810	2.73115	14.525	Yes
DualMean	MLP_H1/mixed	5.26259	28/1728	1.620	6.76038	15.104	Yes
Original RAVM	MLP_H1/clean	5.52908	46/1728	2.662	4.94707	19.502	Yes
Original RAVM	MLP_H1/mixed	5.52045	57/1728	3.299	6.07580	19.618	Yes
Always-FT	MLP_H1/clean	4.95625	648/1728	37.500	137.44362	100.000	No
Always-FT	MLP_H1/mixed	5.10057	740/1728	42.824	254.71456	100.000	No

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
Joint Gate	MLP_H4/clean	4.31795	46/1728	2.662	2.91591	20.660	Yes
Joint Gate	MLP_H4/mixed	4.38438	95/1728	5.498	15.30041	20.949	No
RecentMean	MLP_H4/clean	4.34410	13/1728	0.752	1.62485	14.757	Yes
RecentMean	MLP_H4/mixed	4.33352	88/1728	5.093	17.57693	21.238	No
MWU-Mean	MLP_H4/clean	4.47939	22/1728	1.273	1.86279	14.178	Yes
MWU-Mean	MLP_H4/mixed	4.57145	27/1728	1.562	5.91191	11.921	Yes
DualMean	MLP_H4/clean	4.47609	14/1728	0.810	1.65842	12.326	Yes
DualMean	MLP_H4/mixed	4.49120	33/1728	1.910	6.99313	14.120	Yes
Original RAVM	MLP_H4/clean	4.49646	39/1728	2.257	2.06749	21.296	Yes
Original RAVM	MLP_H4/mixed	4.57237	55/1728	3.183	5.55952	20.775	Yes
Always-FT	MLP_H4/clean	4.00324	568/1728	32.870	76.36033	100.000	No
Always-FT	MLP_H4/mixed	4.11492	707/1728	40.914	162.85507	100.000	No
Joint Gate	TCN_H1/clean	4.39902	6/1728	0.347	0.22704	13.252	Yes
Joint Gate	TCN_H1/mixed	4.40203	53/1728	3.067	6.38022	17.593	Yes
RecentMean	TCN_H1/clean	4.40226	3/1728	0.174	0.10860	10.880	Yes
RecentMean	TCN_H1/mixed	4.32876	67/1728	3.877	8.99640	20.544	Yes
MWU-Mean	TCN_H1/clean	4.52813	0/1728	0.000	0.00000	7.928	Yes
MWU-Mean	TCN_H1/mixed	4.54512	9/1728	0.521	0.71487	9.086	Yes
DualMean	TCN_H1/clean	4.47150	3/1728	0.174	0.10860	10.185	Yes
DualMean	TCN_H1/mixed	4.48659	28/1728	1.620	3.62163	13.600	Yes
Original RAVM	TCN_H1/clean	4.42672	22/1728	1.273	0.55589	29.167	Yes
Original RAVM	TCN_H1/mixed	4.51935	50/1728	2.894	4.15537	25.405	Yes
Always-FT	TCN_H1/clean	3.98117	370/1728	21.412	30.76513	100.000	No
Always-FT	TCN_H1/mixed	4.09578	670/1728	38.773	123.32552	100.000	No
Joint Gate	TCN_H4/clean	5.46950	22/1728	1.273	1.00965	15.741	Yes
Joint Gate	TCN_H4/mixed	5.47065	99/1728	5.729	15.19160	23.438	No
RecentMean	TCN_H4/clean	5.45438	12/1728	0.694	0.65856	13.657	Yes
RecentMean	TCN_H4/mixed	5.39785	120/1728	6.944	20.32175	26.678	No
MWU-Mean	TCN_H4/clean	5.62264	3/1728	0.174	0.12532	10.185	Yes
MWU-Mean	TCN_H4/mixed	5.69074	27/1728	1.562	5.59852	11.574	Yes
DualMean	TCN_H4/clean	5.63124	7/1728	0.405	0.78426	11.285	Yes
DualMean	TCN_H4/mixed	5.69520	38/1728	2.199	5.44583	15.104	Yes
Original RAVM	TCN_H4/clean	5.84604	19/1728	1.100	1.01125	24.248	Yes
Original RAVM	TCN_H4/mixed	5.94131	67/1728	3.877	5.42133	23.785	Yes
Always-FT	TCN_H4/clean	5.01573	381/1728	22.049	38.48548	100.000	No
Always-FT	TCN_H4/mixed	5.14702	669/1728	38.715	137.21130	100.000	No

Table S5: E3: all twelve methods on 40 previously exposed Alibaba machines, three seeds and 24 stages. Risk qualification of Joint does not imply a relative MSE advantage.

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
Joint Gate	clean	2.24203	16/2880	0.556	0.31849	3.090	Yes
Joint Gate	mixed	2.28668	114/2880	3.958	7.66545	7.812	Yes
RecentMean	clean	2.23746	5/2880	0.174	0.30223	1.632	Yes
RecentMean	mixed	2.27532	109/2880	3.785	8.79348	7.986	Yes
MWU-Mean	clean	2.25672	6/2880	0.208	0.04988	1.493	Yes
MWU-Mean	mixed	2.29165	62/2880	2.153	4.12798	3.681	Yes
DualMean	clean	2.24092	4/2880	0.139	0.14292	1.354	Yes
DualMean	mixed	2.26250	43/2880	1.493	2.32144	3.507	Yes
Original RAVM	clean	2.23981	22/2880	0.764	0.15627	6.111	Yes
Original RAVM	mixed	2.28428	187/2880	6.493	3.78823	15.521	No
RAVM-D2B	clean	2.24210	15/2880	0.521	0.11955	4.688	Yes
RAVM-D2B	mixed	2.27200	56/2880	1.944	1.21152	4.826	Yes
Always-FT	clean	2.19892	985/2880	34.201	6.52393	100.000	No
Always-FT	mixed	2.36695	1352/2880	46.944	82.75211	100.000	No
ER	clean	2.21461	1134/2880	39.375	3.44496	100.000	No
ER	mixed	2.30569	1451/2880	50.382	18.48585	100.000	No
EWC	clean	2.19893	984/2880	34.167	6.46252	100.000	No
EWC	mixed	2.36424	1352/2880	46.944	80.81483	100.000	No
Huber-FT	clean	2.23555	1024/2880	35.556	7.61481	100.000	No
Huber-FT	mixed	2.29462	1383/2880	48.021	32.73861	100.000	No
ValidationCommit	clean	2.26406	0/2880	0.000	0.00000	0.000	Yes
ValidationCommit	mixed	2.26406	0/2880	0.000	0.00000	0.000	Yes
Frozen	clean	2.26406	0/2880	0.000	0.00000	0.000	Yes
Frozen	mixed	2.26406	0/2880	0.000	0.00000	0.000	Yes

Table S6: E4: all seven methods per thermal task/stream. UCI: one house, 24 stages; BSE1: 40 stages; BSE2: 38 retained stages. Rooms/seeds/stages are not independent buildings.

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
Joint Gate	BSE1/unmodified	30.39134	32/120	26.667	2920.37382	65.000	No
Joint Gate	BSE1/offset_2C_1h	43.44573	22/120	18.333	9330.89241	59.167	No
Joint Gate	BSE1/dropout_1h	96.26910	25/120	20.833	25985.66779	54.167	No
RecentMean	BSE1/unmodified	31.29912	31/120	25.833	2051.42009	65.833	No
RecentMean	BSE1/offset_2C_1h	40.29649	31/120	25.833	3513.14676	68.333	No

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
RecentMean	BSE1/dropout_1h	103.54738	28/120	23.333	30235.01224	56.667	No
MWU-Mean	BSE1/unmodified	30.20283	27/120	22.500	4308.88489	56.667	No
MWU-Mean	BSE1/offset_2C_1h	44.36193	21/120	17.500	11968.27167	54.167	No
MWU-Mean	BSE1/dropout_1h	50.28756	21/120	17.500	10370.75953	50.833	No
DualMean	BSE1/unmodified	33.34422	9/120	7.500	4577.75017	27.500	No
DualMean	BSE1/offset_2C_1h	35.94845	7/120	5.833	3474.15545	23.333	No
DualMean	BSE1/dropout_1h	43.34315	6/120	5.000	2138.84035	24.167	No
Original RAVM	BSE1/unmodified	33.24402	7/120	5.833	2644.15362	17.500	No
Original RAVM	BSE1/offset_2C_1h	32.17220	5/120	4.167	286.80828	14.167	No
Original RAVM	BSE1/dropout_1h	39.98230	7/120	5.833	781.30368	15.833	No
Always-FT	BSE1/unmodified	29.75514	48/120	40.000	3464.46785	100.000	No
Always-FT	BSE1/offset_2C_1h	50.25128	57/120	47.500	14741.65640	100.000	No
Always-FT	BSE1/dropout_1h	125.89065	63/120	52.500	30853.86974	100.000	No
Frozen	BSE1/unmodified	74.35406	0/120	0.000	0.00000	0.000	Yes
Frozen	BSE1/offset_2C_1h	74.35406	0/120	0.000	0.00000	0.000	Yes
Frozen	BSE1/dropout_1h	74.35406	0/120	0.000	0.00000	0.000	Yes
Joint Gate	BSE2/unmodified	10.67555	22/114	19.298	707.96600	56.140	No
Joint Gate	BSE2/offset_2C_1h	12.64304	16/114	14.035	795.55283	45.614	No
Joint Gate	BSE2/dropout_1h	13.99160	14/114	12.281	781.14040	41.228	No
RecentMean	BSE2/unmodified	10.76604	27/114	23.684	848.55000	57.018	No
RecentMean	BSE2/offset_2C_1h	12.26502	17/114	14.912	777.14685	50.877	No
RecentMean	BSE2/dropout_1h	16.27328	17/114	14.912	1007.87580	42.105	No
MWU-Mean	BSE2/unmodified	10.72833	11/114	9.649	291.47919	37.719	No
MWU-Mean	BSE2/offset_2C_1h	12.22071	14/114	12.281	527.40555	38.596	No
MWU-Mean	BSE2/dropout_1h	14.53977	8/114	7.018	532.44545	31.579	No
DualMean	BSE2/unmodified	11.72182	14/114	12.281	327.59301	38.596	No
DualMean	BSE2/offset_2C_1h	12.72125	5/114	4.386	149.44047	28.070	No
DualMean	BSE2/dropout_1h	14.37846	7/114	6.140	253.32232	26.316	No
Original RAVM	BSE2/unmodified	13.12102	5/114	4.386	98.07257	21.053	No
Original RAVM	BSE2/offset_2C_1h	14.67887	2/114	1.754	2.00759	15.789	Yes
Original RAVM	BSE2/dropout_1h	17.60848	5/114	4.386	472.56529	19.298	No
Always-FT	BSE2/unmodified	10.36885	51/114	44.737	1548.10577	100.000	No
Always-FT	BSE2/offset_2C_1h	14.22358	46/114	40.351	2769.38293	100.000	No
Always-FT	BSE2/dropout_1h	23.51638	36/114	31.579	4263.05345	100.000	No
Frozen	BSE2/unmodified	20.60916	0/114	0.000	0.00000	0.000	Yes
Frozen	BSE2/offset_2C_1h	20.60916	0/114	0.000	0.00000	0.000	Yes
Frozen	BSE2/dropout_1h	20.60916	0/114	0.000	0.00000	0.000	Yes
Joint Gate	uci_T1/clean	87.29398	13/72	18.056	5195.88163	62.500	No
Joint Gate	uci_T1/bias	88.95787	11/72	15.278	8644.93813	59.722	No
Joint Gate	uci_T1/dropout	97.22229	13/72	18.056	7456.54833	62.500	No
Joint Gate	uci_T1/stale	85.93226	12/72	16.667	5667.47310	59.722	No
RecentMean	uci_T1/clean	87.29398	13/72	18.056	5195.88163	62.500	No
RecentMean	uci_T1/bias	88.82141	13/72	18.056	11953.62189	65.278	No
RecentMean	uci_T1/dropout	97.22229	13/72	18.056	7456.54833	62.500	No
RecentMean	uci_T1/stale	86.94845	12/72	16.667	5208.81943	62.500	No
MWU-Mean	uci_T1/clean	84.05331	12/72	16.667	3121.83681	54.167	No
MWU-Mean	uci_T1/bias	80.28544	7/72	9.722	5863.07680	50.000	No
MWU-Mean	uci_T1/dropout	91.39094	11/72	15.278	5574.09314	56.944	No
MWU-Mean	uci_T1/stale	85.27188	11/72	15.278	4678.29677	56.944	No
DualMean	uci_T1/clean	112.60468	6/72	8.333	2604.24685	27.778	No
DualMean	uci_T1/bias	109.52379	5/72	6.944	3919.50736	27.778	No
DualMean	uci_T1/dropout	112.01614	3/72	4.167	487.32225	25.000	No
DualMean	uci_T1/stale	116.11997	6/72	8.333	2521.87218	27.778	No
Original RAVM	uci_T1/clean	126.06785	1/72	1.389	2693.38507	13.889	No
Original RAVM	uci_T1/bias	117.05368	0/72	0.000	0.00000	11.111	Yes
Original RAVM	uci_T1/dropout	124.06467	3/72	4.167	3688.48280	15.278	No
Original RAVM	uci_T1/stale	118.96870	1/72	1.389	2536.66611	13.889	No
Always-FT	uci_T1/clean	103.83556	25/72	34.722	12115.85012	100.000	No
Always-FT	uci_T1/bias	129.40716	24/72	33.333	26744.42858	100.000	No
Always-FT	uci_T1/dropout	118.26630	26/72	36.111	16625.72811	100.000	No
Always-FT	uci_T1/stale	104.42958	23/72	31.944	12035.74066	100.000	No
Frozen	uci_T1/clean	127.83721	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T1/bias	127.83721	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T1/dropout	127.83721	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T1/stale	127.83721	0/72	0.000	0.00000	0.000	Yes
Joint Gate	uci_T2/clean	148.12421	14/72	19.444	12958.32728	54.167	No
Joint Gate	uci_T2/bias	194.18755	6/72	8.333	8330.04040	41.667	No
Joint Gate	uci_T2/dropout	187.17946	11/72	15.278	11675.66272	54.167	No
Joint Gate	uci_T2/stale	142.82572	11/72	15.278	13080.18083	54.167	No
RecentMean	uci_T2/clean	148.14058	14/72	19.444	12699.55429	55.556	No
RecentMean	uci_T2/bias	202.66204	10/72	13.889	10077.40758	51.389	No
RecentMean	uci_T2/dropout	171.66496	11/72	15.278	11675.66272	52.778	No
RecentMean	uci_T2/stale	139.92579	11/72	15.278	12767.97272	55.556	No
MWU-Mean	uci_T2/clean	146.91456	14/72	19.444	11892.75045	54.167	No
MWU-Mean	uci_T2/bias	189.86600	5/72	6.944	8636.94410	40.278	No
MWU-Mean	uci_T2/dropout	191.90592	11/72	15.278	9995.47339	48.611	No
MWU-Mean	uci_T2/stale	139.80327	9/72	12.500	10177.09656	48.611	No
DualMean	uci_T2/clean	217.55356	7/72	9.722	5255.15362	15.278	No
DualMean	uci_T2/bias	278.37437	6/72	8.333	21320.07650	15.278	No
DualMean	uci_T2/dropout	245.61507	8/72	11.111	8366.85850	18.056	No
DualMean	uci_T2/stale	214.14071	1/72	1.389	578.88140	13.889	No
Original RAVM	uci_T2/clean	317.56349	1/72	1.389	782.47534	8.333	No
Original RAVM	uci_T2/bias	389.68703	1/72	1.389	139.68462	6.944	No

Method	Stream	MSE $\times 10^3$	Harm / opp.	Harm (%)	Downside $\times 10^6$	Commit (%)	Qual.
Original RAVM	uci_T2/dropout	410.66744	2/72	2.778	2325.54341	8.333	No
Original RAVM	uci_T2/stale	281.75546	1/72	1.389	578.88140	9.722	No
Always-FT	uci_T2/clean	160.14680	34/72	47.222	45766.27889	100.000	No
Always-FT	uci_T2/bias	228.54924	28/72	38.889	55847.32709	100.000	No
Always-FT	uci_T2/dropout	264.60494	28/72	38.889	75529.21390	100.000	No
Always-FT	uci_T2/stale	168.61441	36/72	50.000	47352.89154	100.000	No
Frozen	uci_T2/clean	613.55787	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T2/bias	613.55787	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T2/dropout	613.55787	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T2/stale	613.55787	0/72	0.000	0.00000	0.000	Yes
Joint Gate	uci_T3/clean	250.42234	16/72	22.222	24503.61673	62.500	No
Joint Gate	uci_T3/bias	300.60238	11/72	15.278	44543.77269	45.833	No
Joint Gate	uci_T3/dropout	315.24485	14/72	19.444	33285.62463	56.944	No
Joint Gate	uci_T3/stale	246.97829	15/72	20.833	28834.91159	61.111	No
RecentMean	uci_T3/clean	249.67904	15/72	20.833	24403.25548	63.889	No
RecentMean	uci_T3/bias	306.81818	14/72	19.444	44945.00372	61.111	No
RecentMean	uci_T3/dropout	315.24485	14/72	19.444	33285.62463	56.944	No
RecentMean	uci_T3/stale	246.97829	15/72	20.833	28834.91159	61.111	No
MWU-Mean	uci_T3/clean	264.59814	11/72	15.278	25521.72026	55.556	No
MWU-Mean	uci_T3/bias	358.14630	6/72	8.333	21738.79587	40.278	No
MWU-Mean	uci_T3/dropout	293.39594	12/72	16.667	19371.27937	48.611	No
MWU-Mean	uci_T3/stale	257.05026	9/72	12.500	16795.07210	48.611	No
DualMean	uci_T3/clean	363.66277	3/72	4.167	12935.84378	34.722	No
DualMean	uci_T3/bias	589.95988	5/72	6.944	2857.29600	30.556	No
DualMean	uci_T3/dropout	571.81317	5/72	6.944	3651.18677	31.944	No
DualMean	uci_T3/stale	355.77297	4/72	5.556	12780.18294	37.500	No
Original RAVM	uci_T3/clean	740.27279	4/72	5.556	24107.61205	22.222	No
Original RAVM	uci_T3/bias	667.91373	5/72	6.944	29642.74839	25.000	No
Original RAVM	uci_T3/dropout	752.66365	4/72	5.556	28503.88834	22.222	No
Original RAVM	uci_T3/stale	743.60524	4/72	5.556	23954.01233	22.222	No
Always-FT	uci_T3/clean	189.22155	18/72	25.000	19876.26360	100.000	No
Always-FT	uci_T3/bias	324.32195	32/72	44.444	75631.23529	100.000	No
Always-FT	uci_T3/dropout	325.58957	29/72	40.278	50555.65636	100.000	No
Always-FT	uci_T3/stale	195.21185	23/72	31.944	22414.79243	100.000	No
Frozen	uci_T3/clean	1478.61656	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T3/bias	1478.61656	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T3/dropout	1478.61656	0/72	0.000	0.00000	0.000	Yes
Frozen	uci_T3/stale	1478.61656	0/72	0.000	0.00000	0.000	Yes

S5 External Inputs, Preprocessing, and Checkpoint Provenance

Alibaba. The provider's 2018 machine-usage trace supplies seven physical utilization fields. Valid CPU targets must be in $[0, 100]$ with a timestamp; records are deduplicated by timestamp, chronologically sorted, and truncated to at most 17,000 valid observations (at least 16,760 required). Known percentage fields are divided by 100. Nonfinite, negative, or over-100 metric values are masked, and the undocumented-scale `mkpi` field is always unavailable. Seven fields are padded to twelve metric slots with zero masks; masks make 24 model inputs. Target is CPU utilization/100. No interpolation is used; irregular physical timestamps are retained although role boundaries use observation indices. The original pool construction excludes earlier exposed machine hashes, ranks eligible machines by SHA-256 of `7311:`*machine-id*, and assigns 20 source, 20 development, 20 calibration, and 40 evaluation machines. The last 40 (identifiers 20060–20099) subsequently became exposed through earlier research; the present E3 does not restore their blind status. Original pretrained GRU sources use the first 20 machines and the same 50-epoch Adam source routine. Existing per-machine warm-up states are reused by recorded identity. The raw archive SHA-256, source/processed-file hashes, deduplication rules, and sorted raw-machine membership are supplied in `E3_ALIBABA_AUTH.json` and `historical_prepare_alibaba.py`.

UCI home. The 19,735 ten-minute records use T1/RH1 through T9/RH9 plus daily sine/cosine (20 values), standardized with mean and population SD from $[0, 4000)$, then concatenated with 20 presence masks. There were no missing observations in that source CSV. The standardized T1/T2/T3 series are separate targets. The original 200-step SGD source fits use $[0, 3000)$; $[3500, 4000)$ supplies a development reference; historical calibration uses $[4000, 9000)$; the new evaluation retains the original final region starting at 9000. Warm-up is $[9000, 9300)$ and history $[9300, 9500)$. At stage t, Update is $[9500 + 360(t-1), 9700 + 360(t-1))$, Verification is the next 160 observations, Service the following 200. Lookback is 96 and horizon 1. The fixed Joint/neighbor gates are transferred from the workload source; no task-specific threshold is selected from these results. Physical MSE equals normalized MSE times the squared source-only target SD. Rooms share one house.

ThermBuild. Only raw measured BSE1/BSE2 data are used. The indoor target is `room_5_living_110_AT`. Ten other inputs are outside air temperature, beam/diffuse horizontal irradiation, indoor heat supply, heat-pump

electrical power and speed, store-to-floor flow temperature, flow volume, and heat/cool modes. Source days 1–120 alone determine each building's observed-value mean and population SD (constant-input SD set to one). Inputs carry a past observed value for at most four samples and otherwise use standardized zero, with original presence masks retained; four daily/annual sine/cosine values make 26 inputs. Targets remain missing when originally missing. Main-file TIME is ordinal and equals temperature-file TIME minus one, not an asserted UTC timestamp. Source fits use 1,000 SGD steps, followed by a development reference on days 121–180; calibration was days 181–270, final days 271–444. Each evaluation has an 800-row warm-up, 200-row initial history, 192-row Update, Verification and Service, stride 384 and $L = 96, H = 1$. The fixed origin-quality checks require at least 48 observed-target origins in Update/Service and at least eight in each of six Verification blocks. Blocks are formed before invalid-target exclusion. No alternate start is searched if initial history fails; no labels are imputed. Final BSE1 retains raw stages 1–40; BSE2 retains 1 and 4–40, excluding 2 and 3 by coverage criteria. Full bounds, retained/excluded counts and exact scales are in the supplied group JSON files.

Asset identity. `SOURCE_CHECKPOINT_INDEX.csv` lists every group's source-file identity (including full SHA-256), seed and actual byte size. The historical source scripts and sidecars describe the available fitting procedures; those are not a newly conducted replication. The separate `training_bundle` now embeds the frozen source/common M0 checkpoint bytes and processed Final inputs for the five submitted synthetic groups; measured-domain input/checkpoint arrays are excluded from this submission subset. A deduplicated asset manifest binds included files to full SHA-256 and size. Original paths in provenance are descriptive; the launcher resolves operational paths inside the attachment. This supplies the lifecycle starting assets without requiring every saved candidate or the author's original drives. One complete synthetic source-fitting path is separately checked in S10; full upstream fitting grids and development selection are not replayed.

S6 Exposure History, Statistical Units, and Measured Cost

The original submission evaluated a recent-and-history RAVM rule. Later D3f development encountered strong neighboring gates; an earlier D3f-2 Confirm stopped under its fixed go/no-go criteria (`STOP_CONFIRM`). The corrected Main cannot be described as continuation of an unbroken positive blind confirmation. A later Smoke initially failed an audit of native legacy clean inputs: an engineering correction set native clean alpha to zero, keeping mixed and coherent streams, gate rules, centers, seeds and valid reusable scientific work unchanged. The failed round used 394,400 optimizer steps; 24,000 were excluded and 24,000 corrective steps executed, reusing 370,400 valid steps. Corrected composite scientific steps were 394,400, while total actual failed-plus-corrective effort was 418,400. Smoke remains development/exposure evidence. Corrected new Main has 3,806,400 actual SGD steps versus nominal cap 4,152,000; none of the present table reconstructions adds optimizer steps. Historical source and failure/correction lock records are supplied separately. The 26/27/8-page October18 PDFs are historical artifacts; their exact editable source correspondence remains unestablished and they are not the true-diff baseline.

Main uses 32 centers as independent units, averaging seeds and stages within center and clean/mixed with equal weights. Coherent-sticky enters risk eligibility, not the balanced MSE average. Source qualification tests each stream's opportunity-based harm and downside, with $\eta = 5.6036475532487204 \times 10^{-5}$; its numerical equality to gate margin δ does not equate their meanings. Center-paired percentile intervals use 4,000 bootstrap resamples, seed 20261020 in Main and 20261022 in supplemental center/machine comparisons. Seeds, stages and windows remain clustered. Intervals do not correct for multiplicity or the Final best-comparator selection. A contrast crossing zero does not demonstrate equivalence/noninferiority. E3 uses 40 machine clusters; E1/E2 use 24 centers. Three rooms and two same-site houses cannot support a building-population interval. No pooled cross-task normalized MSE is meaningful.

All E1–E4 tables retain negative outcomes: Joint mixed20% harm 147/1,728; all tested H4 mixed groups fail the source limit; Alibaba Joint qualifies but loses the MSE comparison to fixed DualMean; all thermal Joint streams fail transferred source qualification. The 13,626 supplemental scientific trajectories and 36 E5 timing replicas are distinct counts. E0 replays 6,912 frozen Joint states without fitting, reproducing every stored Joint action; differences from DualMean/MWU-Mean/RecentMean number 971/699/758. These are action differences on Joint's own state chain, not separate component-causal gains.

E5 has three repetitions of all twelve methods on one fixed center/seed/clean 24-stage lifecycle, one ready lifecycle at a time per repetition. Maintenance runs from entry through first Service access: adaptation, Verification/gate, persistence, sealing/bookkeeping and any native post-seal Fisher work are inside. Shared initial warm-up, later

Service evaluation and late-feedback generation are outside. Joint averages 14.28 s per lifecycle (13.64–14.89); DualMean 11.70 s (11.47–11.92), not seconds per stage. The three-repetition range is not a population confidence interval. The original Main's parallel scheduling totals are not a matched benchmark against E5. CSVs separately retain timing components, Service, feedback, observed process RSS, process GPU peaks and logical buffer bytes. These are not energy measurements, isolated per-stage memory or total system resource use. Environment: Python 3.13.5, PyTorch 2.13.0/CUDA12.6, 20 logical CPUs, approximately 16GB RAM, NVIDIA RTX3060 Laptop 6GB; shared-host contention remains possible.

S7 Method Positioning and Primary-Source Scope

ADWIN [20] updates the retained temporal data window and can trigger model revision or maintain learner statistics. Its means compare two time subwindows, with a significance threshold depending on window contents/lengths and confidence, not a candidate–incumbent deployment conjunction. It requires temporal history. Its change-detection guarantees assume independent bounded observations; contamination of the monitoring/verification channel is not explicitly authenticated. Seldonian regression [21] partitions data into candidate-selection and safety-test sets, using confidence bounds on specified constraint estimators to return a solution or no solution. It provides probabilistic constraint statements under its stipulated assumptions, rather than claiming any statistic is a safety certificate. It is not tied to a historical retention buffer. Neither is rerun as an extra benchmark in this study.

Our RecentMean, MWU-Mean and DualMean are named protocol controls, not external published algorithms: current mean threshold; mean plus a hard MWU cutoff; and current plus historical mean conditions, respectively. Joint changes the deployment state using a continuous current mean–rank trade-off with a separate $d_t < 0$ condition. It accepts potentially unreliable current Verification as evidence and is evaluated empirically. It needs current valid losses, not historical loss as a formula input (shared execution still computes history for controls). Its contribution is the specified trade-off within a causal lifecycle, not invention of MWU or candidate/validation separation. Main supports a bounded empirical gain; it does not identify independent causal benefits of both terms. The ADWIN author preprint sections1/3 and the Science Seldonian paper's framework/Figure1 were read for these comparisons. Two previously cited DT-update papers (Li et al.; Rodriguez et al.) were not used for detailed method claims because their full method content was not verified in this bounded check.

Sharma et al.'s author preprint [26], sections1–2, reports short spikes, noisy intervals and abnormal constant readings, and separately discusses constant calibration offsets. Sinopoli et al. [27], introduction and arrival-process model, studies missing measurements over unreliable channels. These are primary mechanism sources. Neither establishes that this study's .05/.10/.20 fractions, .2 normalized offsets, 2°C episode or one-hour duration matches a particular field installation. Past-only replay models stale information operationally; an incident-frequency or attack-detection claim is outside the evidence here. For quantitative context, the same Sharma author manuscript (p. 4, introduction) reports detected faulty-sample fractions below 0.01% in one six-month dataset, approximately 20% in another three-month dataset, and 15–35% across nodes of a further dataset. These heterogeneous sensor deployments and detector-dependent fractions are not a calibration dataset for our DT tasks. They support testing multiple stress levels, but do not identify our exact fractions, amplitudes or durations as field estimates.

S8 Supplied Materials and Read-Only Reconstruction

The accompanying `08_Reproduction/README.md` gives the runnable aggregation command, input locations and expected outputs. It regenerates the E3 compact table, checks the Main 4.85% result and center wins, independently rebuilds E3 machine-cluster MSE contrast/interval and all method/stream harm frequencies, and verifies E5 lifecycle sums. It also reconstructs plots from the supplied aggregate CSVs. `CLAIM_FILE_PROVIDED.csv` distinguishes actual attachments from indexed assets. The read-only reconstruction entry is separate from the explicitly invoked training entry and bounded engineering acceptance in S10.

The submitted attachments are the Supplement PDF, an independent LaTeX source archive, and `06_Reproducibility_Materials.zip`. The latter contains aggregate CSVs and reconstruction/profile scripts, frozen configurations and recipe descriptions, attribution/identity records, and a bounded synthetic `training_bundle`. Its five unchanged group definitions cover Main, the three accepted architecture/horizon cases and one cost case, with all their referenced processed inputs and frozen source/M0 weights. These assets support sixteen previously accepted

synthetic PATH_A cases plus the specified matching synthetic PATH_B. The historical nineteen-case acceptance also covered Alibaba, UCI and ThermBuild, whose measured input/checkpoint bytes are not in this submitted subset; provider acquisition instructions and recipe descriptions are supplied for them. No claim that all nineteen input paths are self-contained is made. The explicit training entry writes to a new external output directory; verification/preflight performs no optimizer update. The pinned cache is Windows-specific; other operating systems and cross-device numerical identity are not validated. E0 aggregate replay and E5 recorded lifecycle-time sums retain the separate read-only entry; reproduction timings on another machine are not guaranteed to match.

Third-party raw datasets remain obtainable from their original providers [23, 24, 25]; UCI uses CC BY4.0 and ThermBuild CC BY-SA4.0, while Alibaba research/study terms apply. `DATA_ACCESS_AND_TERMS.md` supplies provider links, raw acquisition steps, archive identity, derivative attribution and access boundaries. The submitted descriptive field subsets retain their source-specific notices. Alibaba/UCI/ThermBuild training-input/checkpoint arrays are excluded from this submission; they are not represented as attached. Original ingestion/source-fitting recipes and the frozen source/M0 entry are distinguished; this attachment does not independently rerun the entire upstream development history. Raw archives, all intermediate candidates and large trajectory stores are not necessary inputs to the supplied lifecycle entry and remain omitted. The attachment is delivered for review/research; no public deposit, repository DOI or access-on-request promise is invented.

S9 Post-hoc Monitoring-Quality Evidence

This descriptive profile uses the original fixed evaluation assets and raw observations before filtering, filling or corruption; it does not select conditions by prediction error. The ledger `NUMERICAL_EVIDENCE_LEDGER.csv` records the source, unit, denominator, mapping and unsupported inference for each numerical setting. The associated CSVs give asset/channel counts, the original Update/Verification windows, chronological invalid-run lengths and window quantiles. Model-slot padding is excluded from physical-channel denominators. Unknown-scale `mkpi` is not assigned a percentage-range fault label.

Table S7: Native quality in the already fixed assets. Missing/invalid measurements are not labelled physical fault events or estimates of artificial corruption probability.

Scope	Raw evidence and denominator
Alibaba, 40 fixed machines	2,457,461 selected raw records from a complete archive scan; CPU has one nonfinite value; `mem_gps` has 1,952,327 nonfinite values; `disk_io_percent` has 932 explicit out-of-range markers.
Alibaba role windows	All 1,920 original Update/Verification windows have a 100% any-invalid-row fraction, driven by persistent unavailable `mem_gps`; this is not a 100% all-sensor fault rate.
ThermBuild BSE1/BSE2	42,528/42,483 raw rows; target missing counts 605/462. Of 156 retained role windows, 152 have no invalid row and four have 4/192 (2.0833%).
UCI home	19,735 raw records; no nonfinite numeric observations.

Alibaba records were chronologically sorted before run-length calculation. Modal interarrival intervals describe observed timestamp regularity, not a certified schedule; interval deviations cannot distinguish acquisition loss from machine inactivity. Main-file and room-temperature clocks in ThermBuild differ by one index; measured cadence is 15 minutes, without an asserted UTC origin. Longer raw thermal gaps, including 223 samples (55.75 hours) within the evaluation input extent, are reported separately from admissible role windows and do not estimate one-hour fault incidence. The original quality exclusions and causal preprocessing remain unchanged.

The frozen 5/10/20% fractions, normalized amplitude .2, six-observation stale record and one-hour episode remain *stress settings*. For percentage utilization divided by 100, .2 is 20 percentage points, not 20% relative error. Temperature offsets must instead be divided by each source-only SD: a 2°C offset corresponds to approximately 2.13136 and 1.83461 standardized units in BSE1 and BSE2. Granderson et al. [28], Table 5 and Figure 7, include $\pm 2/\pm 4$ °C zone-sensor biases and a +2°C example. The fan-coil subsystem uses HVACSIM+ simulation; these are controlled benchmark magnitudes, not observed incident rates at our site. Native unavailable channels support a missing-observation mechanism, not injected bias prevalence. Consequently site-specific calibration of the full fractions/amplitudes/durations remains unresolved. The predeclared optional new-condition test was skipped because its mapping/novelty conditions were not all met; no additional scientific condition or optimizer step was introduced for this profile.

Public field-data cross-check. We obtained the three version-3 operational AHU CSVs associated with Wang [29], verifying the provider's file sizes/MD5 and sealing SHA-256. They contain 113,375 auditorium, 66,048 hospital and 175,532 office records. Combining only the provider's return/supply-temperature fault labels gives 2,138, 8,844 and 8,030 rows, respectively: 1.8858%, 13.3903% and 4.5747% of released rows. These are curated hourly operational-state labels, not independent incidents, measured additive bias against a calibrated truth, or our artificial corruption fractions. Original timestamp duplicates and version-specific dates are reported without silent cleaning. Released missing values can include inapplicable seasonal channels. Thus these real records strengthen application context but do not retrospectively calibrate our fractions, offsets or episode durations. The official LBNL RTU inventory additionally documents physically imposed $\pm 2/\pm 4\,^{\circ}\mathrm{C}$ one-day sensor-bias tests; its field subset concerns compressor control and refrigerant charge, not sensor-corruption prevalence. The unit/source/support audit and runnable field-profile entry are supplied.

Paired-reference field measurements. We additionally profiled the released QUANT v1 data [30], selecting all NO_2 systems and all products before calculating discrepancies. The original sensor/reference files passed provider MD5 checks. The selected files contain 751,303 sensor rows and 58,615 ratified-reference rows across London, Manchester and York. Exact city/hour pairing yields 591,316 finite sensor–reference pairs across 119 of 124 system/product strata from 55 systems; five strata have no finite pair and remain reported. No calibration model was fitted. Nonempty manufacturer flags in 5,373 finite pairs were retained, not silently excluded.

Table S8: External paired-reference descriptive evidence. Products share sensors, hours and reference instruments; pair counts are not independent replication counts. Error is sensor minus ratified reference in ppb, not degrees Celsius or utilization. Ranges are across paired system/product strata, not confidence intervals.

Product	Paired/all strata	Pairs	Min. pairs	Max. pairs	Median absolute error range (ppb)
out-of-box	48/49	149,229	181	18,751	1.329–16.276
cal1	43/46	249,481	777	14,857	1.183–28.511
cal2	28/29	192,606	396	19,517	1.842–34.549

This adds observed discrepancies against a co-located reference rather than operational fault labels alone; the reference retains its own uncertainty. Manufacturer-calibrated products are not a new calibration performed here. Full system-level signed/absolute-error quantiles, coverage counts and availability-run descriptions accompany the relative-path read-only entry in `08_Reproduction/field_quant`. Availability runs are bounded by first/last released records and can include maintenance or relocation, so they are not verified fault durations. Hourly averaging cannot resolve sub-hour events. Because this is air-quality monitoring, ppb errors cannot calibrate our temperature/utilization amplitudes, 5/10/20% injected fractions, replay frequency or one-hour incident prevalence. The target-site numerical-calibration limitation remains; no RAVM evaluation case, gate setting or optimizer update was added.

S10 Bounded Reproduction and Assistive Tools

Frozen-input lifecycle path (PATH_A). The synthetic subset supplies actual processed inputs and source/M0 checkpoint bytes, original scientific functions, frozen group specifications and relative-path commands. This is distinct from the broader historical engineering acceptance summarized below. Before acceptance, identities, cases, tolerances and a 50,000-step ceiling were locked. The bounded selection was one Main center (63000), seed 7311 and mixed stream across all twelve methods; Joint in three architecture/horizon cases (GRU H4, MLP H1, TCN H4); Alibaba machine 20060; UCI T1 bias; BSE1 one-hour dropout; and one clean cost case. It is an engineering coverage check, not an independently sampled scientific replication or an expansion of the paper's statistical units.

All 19 full lifecycles passed the original independent audit, covering 472 stages and 22,400 successful optimizer updates. Compared with the corresponding original frozen records, commitment mismatches and model-state-hash mismatches were zero, and maximum absolute MSE difference was zero in the exercised environment. Read-access instrumentation recorded no reads from original-project assets in acceptance children. The provided package supplies the required inputs; the interpreter/runtime remains an external prerequisite. Windows, Python 3.13.5, PyTorch 2.13.0+cu126 and CUDA 12.6 were exercised. These results do not promise identity across operating systems, library versions or devices.

Source-fitting path (PATH_B). Source inputs/recipes, raw acquisition/preparation entries, environment locks, checkpoint identities and a run matrix are supplied separately. The final specified synthetic GRU/H1/seed7311 path uses the original source-training stack (Python 3.11.15, PyTorch 2.5.1+cu121, CUDA12.1; exercised NumPy2.4.6), followed by the recorded Main stack (Python 3.13.5, PyTorch 2.13.0+cu126, CUDA12.6). From freshly extracted assets, source fitting ran all50 epochs/14,150 Adam updates on the original twenty source prefixes, followed by a fresh50-update common target warm-up and all24 Main center63000/mixed/Joint stages (1,200 SGD updates). The new source and M0 state hashes matched their original counterparts. Across all24 stages, all72 parent/candidate/deployed state comparisons and every commitment action matched; maximum absolute MSE difference was zero. The original independent audit passed. State/tensor identity is distinguished from serialized checkpoint-container byte identity. No original-project input or CAS read was allowed; temporary runtime files stayed inside the isolated directory.

This is executed acceptance of one specified PATH_B using the original stage-specific stacks, not an all-domain/source-grid validation, development-selection replay or independent external replication. The earlier two-update source check is only an entry check. Provider data rights remain applicable; no public deposit is claimed. Two declared interpreter environments are runtime prerequisites, not hidden original-project input dependencies. Actual commands, versions and receipts accompany the assets.

Detailed engineering acceptance and attempt-accounting records accompany the technical attachment. These checks do not add scientific samples or replace Main/E0–E5 results.

Assistive-tool provenance. The final framework schematic in Main Figure 2 was redrawn as native vector artwork with OpenAI Codex (client 0.163.0-alpha.11) assisting explicit figure-code preparation. Matplotlib 3.10.0 renders the editable source as vector PDF/SVG and a 1200 dpi RGB TIFF. Its content was checked against the frozen rule and information boundaries; experimental plots were rendered from the sealed CSVs with Python. OpenAI ChatGPT and Codex assisted the software/evidence/analysis-workflow and writing tasks described in the main Methods. The superseded raster version of Figure 2 used OpenAI image generation through Codex; its image-model version was not retained. That historical gap is not filled from a tool nickname or the text-model identifier used for the new vector-code preparation. Recoverable Codex client versions are 0.163.0-alpha.10.2 and 0.163.0-alpha.11; recovered session model identifiers include `gpt-6.1-sol`, `gpt-6-astra` and `gpt-6-sol`. Client version and model identity are distinct. Historical ChatGPT web versions could not be recovered. Deterministic scripts and the documented audits generated/checked reported numerical artifacts; these checks do not substitute for authors' final factual verification and responsibility.

References

[1] Feng B, Ding Z. Application-Oriented Cloud Workload Prediction: A Survey and New Perspectives. Tsinghua Science and Technology. 2025;30(1):34-54.

[2] Kim YM, Song S, Koo BM, Son J, Lee Y, Baek JG. Enhancing Long-Term Cloud Workload Forecasting Framework: Anomaly Handling and Ensemble Learning in Multivariate Time Series. IEEE Transactions on Cloud Computing. 2024;12(2):789-99.

[3] Tao F, Zhang H, Liu A, Nee AYC. Digital Twin in Industry: State-of-the-Art. IEEE Transactions on Industrial Informatics. 2019;15(4):2405-15.

[4] Mihai S, Yaqoob M, Hung DV, Davis W, Towakel P, Raza M, et al. Digital Twins: A Survey on Enabling Technologies, Challenges, Trends and Future Prospects. IEEE Communications Surveys & Tutorials. 2022;24(4):2255-91.

[5] Amad H, Astorga N, van der Schaar M. Continuously Updating Digital Twins Using Large Language Models. In: Proceedings of the 42nd International Conference on Machine Learning. vol. 267 of Proceedings of Machine Learning Research; 2025. p. 1343-66.

[6] Bang J, Koh H, Park S, Song H, Ha JW, Choi J. Online Continual Learning on a Contaminated Data Stream With Blurry Task Boundaries. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition; 2022. p. 9275-84.

[7] Parisi GI, Kemker R, Part JL, Kanan C, Wermter S. Continual Lifelong Learning with Neural Networks: A Review. Neural Networks. 2019;113:54-71.

[8] Wang L, Zhang X, Su H, Zhu J. A Comprehensive Survey of Continual Learning: Theory, Method and Application. IEEE Transactions on Pattern Analysis and Machine Intelligence. 2024;46(8):5362-83.

[9] Ding M, Ji K, Wang D, Xu J. Understanding Forgetting in Continual Learning with Linear Regression. In: Proceedings of the 41st International Conference on Machine Learning. vol. 235 of Proceedings of Machine Learning Research. PMLR; 2024. p. 10978-1001.

[10] Xu H, Wu J, Pan Q, Guan X, Guizani M. A Survey on Digital Twin for Industrial Internet of Things: Applications, Technologies and Tools. IEEE Communications Surveys & Tutorials. 2023;25(4):2569-98.

[11] Qin B, Pan H, Dai Y, Si X, Huang X, Yuen C, et al. Machine and Deep Learning for Digital Twin Networks: A Survey. IEEE Internet of Things Journal. 2024;11(21):34694-716.

[12] Li J, Guo S, Liang W, Wang J, Chen Q, Zeng Y, et al. Digital Twin-Enabled Service Provisioning in Edge Computing via Continual Learning. IEEE Transactions on Mobile Computing. 2024;23(6):7335-50.

[13] Rodriguez F, Chicaiza WD, Sanchez A, Escano JM. Updating Digital Twins: Methodology for Data Accuracy Quality Control Using Machine Learning Techniques. Computers in Industry. 2023;151:103958.

[14] Shen Z, Sun P, Arrano-Vargas F, Konstantinou G. Anomalous Data Evaluation for Reliable Updates of Trustworthy Power System Digital Twins. Expert Systems with Applications. 2026;314:131434.

[15] Gama J, Zliobaite I, Bifet A, Pechenizkiy M, Bouchachia A. A Survey on Concept Drift Adaptation. ACM Computing Surveys. 2014;46(4):1-37.

[16] De Lange M, Aljundi R, Masana M, Parisot S, Jia X, Leonardis A, et al. A Continual Learning Survey: Defying Forgetting in Classification Tasks. IEEE Transactions on Pattern Analysis and Machine Intelligence. 2022;44(7):3366-85.

[17] Kirkpatrick J, Pascanu R, Rabinowitz N, Veness J, Desjardins G, Rusu AA, et al. Overcoming Catastrophic Forgetting in Neural Networks. Proceedings of the National Academy of Sciences. 2017;114(13):3521-6.

[18] Rolnick D, Ahuja A, Schwarz J, Lillicrap T, Wayne G. Experience Replay for Continual Learning. In: Advances in Neural Information Processing Systems. vol. 32; 2019. p. 348-58.

[19] Huber PJ. Robust Estimation of a Location Parameter. The Annals of Mathematical Statistics. 1964;35(1):73-101.

[20] Bifet A, Gavaldà R. Learning from Time-Changing Data with Adaptive Windowing. In: Proceedings of the 2007 SIAM International Conference on Data Mining; 2007. p. 443-8.

[21] Thomas PS, Castro da Silva B, Barto AG, Giguere S, Brun Y, Brunskill E. Preventing undesirable behavior of intelligent machines. Science. 2019;366(6468):999-1004.

[22] Mann HB, Whitney DR. On a Test of Whether one of Two Random Variables is Stochastically Larger than the Other. The Annals of Mathematical Statistics. 1947;18(1):50-60.

[23] Alibaba Open Cluster Trace Program. Alibaba Cluster Trace v2018. Alibaba Open Cluster Trace Program; 2018. Available at: https://github.com/alibaba/clusterdata/blob/master/cluster-trace-v2018/trace_2018.md. Accessed 24 October 2026.

[24] Candanedo L. Appliances Energy Prediction. UCI Machine Learning Repository; 2017. DOI: 10.24432/C5VC8G. Available at: https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction. Accessed 24 October 2026.

[25] Kersken M, Raisch F, Male M, Tischler B. ThermBuild: Real-world and Simulated Thermal Data from 960 Residential Multi-zone Buildings in Central Europe. Fordatis, Fraunhofer-Gesellschaft; 2026. DOI: 10.24406/fordatis/445. Measured raw-data subset only in this study. Available at: https://fordatis.fraunhofer.de/handle/fordatis/486. Accessed 24 October 2026.

[26] Sharma AB, Golubchik L, Govindan R. Sensor faults: Detection methods and prevalence in real-world datasets. ACM Transactions on Sensor Networks. 2010;6(3):1-39.

[27] Sinopoli B, Schenato L, Franceschetti M, Poolla K, Jordan MI, Sastry SS. Kalman Filtering With Intermittent Observations. IEEE Transactions on Automatic Control. 2004;49(9):1453-64.

[28] Granderson J, Lin G, Chen Y, Casillas A, Wen J, Chen Z, et al. A labeled dataset for building HVAC systems operating in faulted and fault-free states. Scientific Data. 2023;10:342.

[29] Wang S. Real operational labeled data of air handling units from office, auditorium, and hospital buildings. Scientific Data. 2025;12:1481.

[30] Diez S, Lacy S, Urquiza J, Edwards P. QUANT: a long-term multi-city commercial air sensor dataset for performance evaluation. Scientific Data. 2024;11:904.